Scene-Aware Radio Evaluation for Fire-Front Mesh Sensing: A Reproducible Sionna RT Integration

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***Abstract*—This paper extends the supplied 2026ESWContest_free_HWAJUNGSAENGRYEON project, whose baseline already includes node-stamped event evidence, silent-node verification, fire-front estimation, and route rebuilding. The added work integrates Sionna RT and compares scene-aware mesh links with distance-only and statistical path-loss baselines. In a fixed 4-by-4 benchmark, the synthetic-forest graph had 34 usable links versus 42 in the open case, while mean delivery was 95.1% versus 96.9% over ten seeds. A separate 1,620-run study found no consistent delivery change across 3-by-3, 4-by-4, and 5-by-5 grids. Adding the baseline's wind model lowers delivery far more than the forest does, with no detectable interaction. All results are synthetic: vegetation is uncalibrated, links use a gain threshold rather than a validated packet-error curve, and each channel is static within a run.**

***Index Terms*—Wildfire monitoring, wireless sensor networks, mesh networking, last-gasp telemetry, event timestamping, Sionna RT, route resilience.**

I. Introduction

A conventional monitoring view treats a sensor that stops communicating as a missing device. In a wildfire, however, silence can occur precisely because the fire has reached a node whose measurements are valuable for estimating the front. A useful response must preserve the node's final evidence, distinguish a fire-caused loss from a temporary radio outage or unrelated failure, and keep reports moving through surviving paths. Prior work has explored priority routing for fire alerts and data sent shortly before a sensor is destroyed by fire [1]; this project focuses on combining timestamp-aware death evidence with explicit mesh-channel evaluation.

The practical questions addressed are: (1) Why use a wireless mesh for this monitoring task? (2) What is a mesh sensor network? (3) Where is it useful? (4) How does this project use it to preserve information and estimate fire-front motion? (5) What did the implementation and simulation show, and what remains unverified?

This article explains the wireless mesh concept and its use in remote hazard monitoring, then reports the project workflow, Sionna integration, experiments, and limits. The original sensor-death timestamping, verification, estimation, and route-repair logic is part of the supplied baseline; the new work evaluated here is the channel-model integration and its reproducible comparison. The five practical questions are summarized after the results.

II. Related Work and Research Position

Wireless sensor networks have been studied for continuous environmental monitoring and fire-alert delivery. Research on wildfire detection includes multi-sensor deployment, energy-aware routing, risk classification, and alert dissemination. Such work establishes that sensing quality and network lifetime are coupled: when a path relies on nodes exposed to the same hazard, an apparently valid route can disappear as the event develops. Koga *et al.* studied priority routing for forest-fire alerts, including high-priority data sent shortly before node destruction [1]. More recent hazardous-event work evaluates redundant paths and recovery after multiple node failures [2]. These studies show that urgent delivery and route recovery are established research directions; the present work does not claim either as a new protocol. It evaluates how the radio graph supplied to an existing mesh simulator changes under a scene-based channel.

For radio modeling, log-distance path loss and log-normal shadowing provide useful statistical comparators, while ITU-R P.833 describes vegetation attenuation models whose use depends on frequency and vegetation conditions [3]. Sionna RT instead traces paths through explicit geometry and material parameters [4]. Packet reception also depends on the physical layer and interference; IEEE 802.15.4 studies derive packet-reception behavior for specific PHY assumptions [5]. Because the original simulator does not specify a transceiver, receiver sensitivity, or validated PHY curve, this study reports gain-based topology and does not claim calibrated SNR-to-PER results. A future hardware-specific version should configure a link budget and a measured or standard-matched packet-error curve.

Node-local event timestamps are only useful when clock offset and drift are bounded; reference-broadcast synchronization is one established WSN approach, but this work neither implements nor validates a hardware clock protocol [6]. The contribution here is narrower: it connects channel-model studies to an existing fire-front mesh workflow and makes the path-gain-to-graph comparison reproducible. It is a simulation integration study, not a new death-verification algorithm, routing protocol, or field-validated prediction system.

III. System Model and Performance Definitions

A. Sensor and fire-front model

The mesh contains a finite set of nodes $\mathcal{V} = \{1, \ldots, N\}$ placed at coordinates $\mathbf{p}_i = (x_i, y_i)$. Each node observes a

temperature history and reports measurements with its own timestamp. The reference simulator has a prescribed fire-front direction and speed, which define a ground-truth arrival time for each location. For a locally planar front moving at constant velocity over a limited region, the arrival-time field can be approximated by

$$T(x, y) = ax + by + c + \epsilon, \tag{1}$$

where a and b describe the spatial gradient, c is an offset, and ϵ captures measurement and model error. Given verified event samples (x_i, y_i, t_i), the least-squares plane fit estimates a, b, and c. The gradient $\nabla T = (a, b)$ points toward increasing arrival time. If the front propagates in unit direction $\mathbf{n}$ at constant speed v, then $T(\mathbf{x}) = (\mathbf{n} \cdot \mathbf{x} - c_0)/v$ and $\nabla T = \mathbf{n}/v$. Thus the propagation direction is $+\nabla T/\|\nabla T\|$, consistent with the estimator implementation and tests. Under this planar approximation, the estimated speed is

$$\hat{v} = \frac{1}{\sqrt{a^2 + b^2}}, \tag{2}$$

provided the gradient is nonzero. The estimated arrival time at a target coordinate (x_q, y_q) is $\hat{T}_q = ax_q + by_q + c$. The model is local: curved fronts, changing spread rate, wind shifts, or sparse and clustered event samples can violate the planar approximation.

B. Communication graph

At time t, the network is represented as a graph $G_t = (\mathcal{V}_t, \mathcal{E}_t)$ over the surviving nodes. A candidate edge between nodes i and j is enabled only when the pair satisfies both the radio-range constraint and the bidirectional gain constraint:

$$(i, j) \in \mathcal{E}_t \iff d_{ij} \le R \quad \wedge \quad \min\{g_{ij}, g_{ji}\} > \tau, \tag{3}$$

where R is the configured range, g_{ij} and g_{ji} are path gains in dB, and τ is the minimum link threshold. The minimum of the two directional gains is used because a path must support communication in both directions in this model. Once a fire death is verified, the dead node is removed and routes are rebuilt over G_t.

C. Event-time integrity and verification

The event timestamp and the gateway's confirmation timestamp refer to different events. Let t_i^{node} be the timestamp recorded by node i, and let t^{gw} be the time when the gateway receives or confirms the report. Substituting t^{gw} for t_i^{node} introduces a network-dependent delay $\delta_i = t^{gw} - t_i^{node}$. If δ_i varies by node, the spatial ordering of arrival samples can be biased. The project adapter therefore uses the node's last-gasp timestamp when available and falls back to the node-stamped heartbeat when it is not; gateway confirmation time is not treated as event time.

A silent node is not automatically accepted as a fire-front sample. The verifier waits for a configured silence interval and requires independent neighbor evidence (with the configured confirmation count capped by the node's available neighbor count). It then evaluates temperature and spatial consistency, separating fire-related losses from non-fire failures. This check reduces the risk that a temporary communication gap or an unrelated failure is interpreted as a front-arrival observation. It does not eliminate false alarms under all possible correlated failures, which must be addressed by additional field testing.

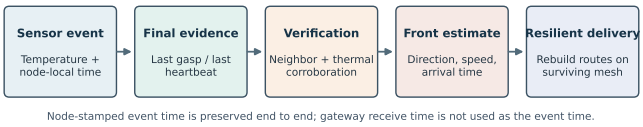


Fig. 1. End-to-end event and communication workflow. The event timestamp remains attached to the sensor evidence as it passes through verification and estimation.

D. Reported metrics

Final packet-delivery rate is the fraction of generated packets delivered by the experiment's stopping time. Direction error is the angular difference between estimated and reference fire-front directions. Relative speed error is reported as a percentage. Arrival-time error is the absolute error at the modeled target location in seconds. Route-repair delay is measured from confirmed node loss to a repaired route. False-positive rate is the fraction of surviving nodes incorrectly confirmed as fire deaths. These metrics describe different aspects of performance and should not be collapsed into one score: a network may deliver most packets yet produce a biased front direction if event samples are sparse or systematically delayed.

IV. Original-Project Gap and Addressed Problem

The original simulator contains the central event-processing logic: node-stamped last-gasp/heartbeat evidence, multi-neighbor silence confirmation, non-fire rejection rules, estimation from accepted fire deaths, and route rebuilding after confirmed losses. Tests also check that the gateway-confirmation time does not replace the node's event timestamp. These features are retained; this paper does not claim to have newly invented or rewritten them.

The modeling gap addressed by this implementation is in the radio environment used to test the system. A distance-only graph can declare a link usable whenever two nodes are close enough, even if trunks and vegetation would attenuate or block the radio path. The original model therefore cannot quantify how a more structured propagation environment changes the mesh topology and delivery metrics. We added a separate Sionna RT integration that computes a directed path-gain matrix from a 3-D radio scene and feeds it into the existing network configuration. The change is an experimental channel-model extension, not a change to the core death-time estimator.

The baseline was developed and tested under idealized conditions. It already models wind on the fire side (an elliptical front and its length-to-breadth ratio) but not the canopy that surrounds the sensors, so its radio links ignore vegetation. The question addressed here is therefore how the baseline behaves under trees and wind together: trees through a scene-derived channel, and wind through the baseline's own fire-shape model.

The end-to-end design addresses five failure modes:

1) **Information loss at destruction:** retain the last received measurements and accept a node-stamped last-gasp report when available.
2) **Presence-only alarms:** fit a local arrival-time plane from confirmed fire-death observations to estimate direction and speed; project that model to estimate arrival time (ETA).
3) **Network damage:** rebuild routes after confirmed node losses so surviving paths can carry traffic.
4) **False deaths:** require corroboration from neighboring observations and classify likely non-fire losses separately rather than treating every silence as a fire-front point.
5) **Network-delay contamination:** use the node's event timestamp (falling back to its last heartbeat timestamp when the final report is absent) rather than the later gateway confirmation time.

V. How the Implemented Workflow Operates

The node records a local timestamp with each event. When its thermal state reaches the configured dying condition, the simulator/adapter can preserve a last-gasp event before the node becomes unavailable. A gateway adapter maps node-local time to a relative event time and records the time source; if the last-gasp message was lost, it uses the last node-stamped heartbeat. Gateway receive or confirmation time is not substituted for the event time.

A verifier then evaluates whether silence represents a confirmed loss and whether the loss is consistent with fire evidence. It uses neighbor observations and temperature evidence, and excludes cases classified as non-fire from the fire estimator. Accepted death locations and event times form samples for a local arrival-time plane. For the fitted plane $T(x, y) = ax + by + c$, propagation direction is the positive normalized gradient $+\nabla T/\|\nabla T\|$, and speed is $1/\|\nabla T\|$, as verified against the configured fire direction in the baseline tests. The implementation also reports arrival-time error against the reference fire. Following confirmation, the network recomputes paths on the surviving graph. This staged design separates (i) event-time integrity, (ii) death verification, (iii) fire-motion estimation, and (iv) packet routing.

VI. Sionna RT Integration and Experimental Setup

Sionna RT is a ray-tracing tool for modeling radio propagation in three-dimensional scenes; it computes propagation quantities from scene geometry and radio-material properties [4]. It does not model wildfire spread. We generated two scenes: an empty open reference and a simplified forest containing a jittered grid of trunk and canopy obstacles with illustrative vegetation radio properties. The forest is a controlled synthetic geometry, not a measured or botanically calibrated environment.

The fixed layout contains 16 nodes on a 4-by-4 grid with 10 m spacing and 1.5 m antenna height. The carrier frequency is 2.4 GHz. Sionna stores receiver rows and transmitter columns in the simulator's column-major order, $i = x_{cell}\, N_{rows} + y_{cell}$. A candidate edge must satisfy the 18 m range cap and the bidirectional gain threshold. The added model comparison uses a -100 dB gain cutoff. For interpretation only, a 0 dBm transmitter and -100 dBm receiver sensitivity imply this gain cutoff; free-space loss at 10 m and 2.4 GHz is about 60.05 dB, or roughly 40 dB of nominal margin. No radio is specified, so these are illustrative link-budget values, not calibrated hardware settings.

The original fixed benchmark used ten seeds for distance-only, Sionna-open, and three forest-threshold labels (50 runs). A separate ten-seed comparison evaluated five distinct models (50 runs): distance-only, log-distance, log-distance-plus-shadowing, Sionna open, and Sionna forest. The statistical baselines use an illustrative, uncalibrated path-loss exponent of 2.7 and 6 dB log-normal shadowing. Sionna ran on CPU through LLVM 18.1.6 because the available NVIDIA driver did not support the CUDA runtime expected by the installed Sionna version.

VII. Results

In the original -95/-100/-105 dB sweep, every cutoff retained 34 forest links and 95.1% delivery. In the added comparison, log-distance and log-distance-plus-shadowing retained 42 range-limited links and matched distance-only delivery under the selected 18 m cap and -100 dB cutoff. The forest graph retained 34 links. Its paired delivery change was -1.73 percentage points (95% CI [-4.80, 1.34]); a post-hoc TOST with a ±2-point margin did not establish equivalence. The ten-seed forest result is therefore imprecise: it neither demonstrates a robust delivery penalty nor proves negligible impact.

The Sionna open scene reproduces the distance-only graph in this setup. The synthetic forest removes eight links, but the ten-seed delivery interval remains wide. Figure 2 visualizes the derived graph; it is not a field deployment.

VIII. Implementation Architecture and Reproducibility

The implementation preserves a strict boundary between the original simulator and the new radio-scene integration. The existing node model generates temperature and state transitions. The network layer builds a graph, routes traffic, records received heartbeats and temperature reports, and removes confirmed dead nodes from active routes. The verifier consumes silence and thermal evidence and decides whether a death should contribute to fire estimation. The estimator fits the arrival-time surface from accepted samples and reports direction, speed, and arrival-time estimates. The new Sionna scripts sit outside these modules: one generates scene-derived gain matrices and metadata, and a plotting script converts those matrices into a visual mesh topology. This add-on boundary keeps the original estimator and verification behavior reproducible and makes it possible to compare radio assumptions without changing the estimator itself.

The generator creates an empty Sionna scene for the open condition. For the forest condition, it creates repeatable geometric obstacles from a deterministic seed: scaled scene spheres represent trunks and canopy volumes, and sensor positions are kept outside an exclusion radius so that the experiment measures paths through the modeled vegetation rather than antennas embedded inside an obstacle. A vegetation

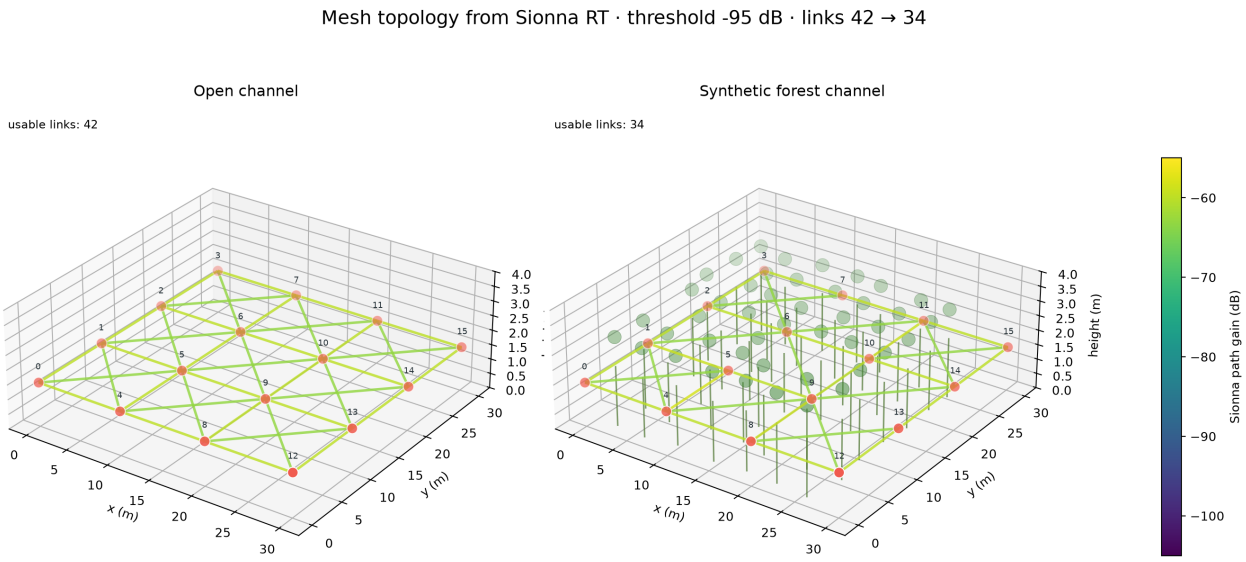


Fig. 2. Three-dimensional mesh topology derived from Sionna RT path-gain matrices. Left: open scene with 42 usable links. Right: simplified synthetic forest with 34 usable links at the -95 dB display threshold. Edge colors encode path gain; nodes and vegetation obstacles are synthetic.

TABLE I
FIXED 4-BY-4 CHANNEL-MODEL COMPARISON. DELIVERY IS MEAN PERCENT WITH A MARGINAL 95% CI; PAIRED CHANGE IS RELATIVE TO DISTANCE-ONLY.

Model	Links	Delivery mean (95% CI)	Paired change (95% CI, pp)	TOST, ±2 pp
Distance-only	42	96.86 [94.97, 98.75]	Reference	–
Log-distance	42	96.86 [94.97, 98.75]	0.00 [0.00, 0.00]	Equivalent
Log-distance + shadowing	42	96.86 [94.97, 98.75]	0.00 [0.00, 0.00]	Equivalent
Sionna open	42	96.86 [94.97, 98.75]	0.00 [0.00, 0.00]	Equivalent
Sionna synthetic forest	34	95.13 [92.72, 97.54]	-1.73 $[-4.80,\ 1.34]$	Not established

TOST uses a post-hoc ±2 percentage-point margin, so it is a sensitivity check, not a preregistered equivalence claim. For the forest comparison, one-sided p values were 0.424 and 0.011; equivalence was not established. The identical baseline-model curves reflect the selected 18 m cap and -100 dB gain cutoff, not universal equivalence of these propagation models.

radio material is assigned to the synthetic objects. The network matrix contains one path-gain value for each ordered transmitter-receiver pair. Diagonal entries represent self-links and are not used to create network edges. Node coordinates are saved beside each matrix and the JSON metadata records scenario name, frequency, grid size, spacing, height, seed, ray depth, units, and matrix ordering.

A consequential integration detail is matrix ordering. The existing simulator maps grid coordinates to a flattened matrix index using $i = x_{cell}N_{rows} + y_{cell}$. The generator uses the same order. A matrix can have the correct 16-by-16 shape and still be wrong if its node order differs; such a mismatch silently assigns channel gains to the wrong physical pairs. A regression test of the matrix contract should therefore verify both dimensions and coordinate-to-index mapping whenever the grid or export format changes.

For the tested workstation, Sionna RT used the CPU LLVM backend through Dr.Jit and LLVM 18.1.6. The available NVIDIA driver did not provide a compatible CUDA backend for the installed Sionna version, so all reported radio-scene generation used CPU execution. This is a backend choice, not a difference in the conceptual channel model. Reproduction requires the Python packages listed in the integration requirements, the LLVM shared library path on Windows, and the same scene seed and command options. The project test suite was run from the isolated environment and completed with 60 passing tests; no source file in the original project core was changed for this integration.

IX. EXPERIMENTAL PROTOCOL

A. Scenario matrix

The original fixed benchmark has five labels and ten seeds (50 runs): distance-only, Sionna open, and three forest cutoffs. A separate ten-seed comparison adds log-distance and log-distance-plus-shadowing baselines. A third, 90-run threshold stress test moves the gain cutoff from its assumed link-budget value toward the observed gain range. Two supplementary checks (30 signed-speed runs and a 60-run last-gasp ablation) are described in the appendix.

B. Execution sequence

The reproducible workflow has four stages. First, the radio scenes are generated and saved as gain matrices with node

TABLE II
SIMULATION AND IMPLEMENTATION SETTINGS

Setting	Value used	Role in experiment
Grid	4×4 (16 nodes)	Sensor placement
Grid spacing	10 m	Horizontal separation between adjacent grid cells
Node height	1.5 m	Radio endpoint height in the Sionna scene
Carrier frequency	2.4 GHz	Scene propagation frequency
Radio range	18 m	Existing simulator's maximum geometric edge length
Path-gain threshold	-100 dB	Illustrative 0 dBm transmitter and -100 dBm sensitivity; stress cutoffs swept separately
Sionna ray depth	2	Maximum modeled path interactions configured by the generator
Forest seed	42	Repeatable obstacle placement for the channel matrix
Experiment seeds	0–9	Ten stochastic simulator runs for each scenario label
Silence timeout	3.0 s	Duration before silence can be considered for verification
Neighbor confirmations	$K = 3$	Required independent observations, capped by available neighbors
Dying-state allowance	0.3 s	Configured time budget for a final report before death
Temperature warning level	60 °C	Thermal evidence threshold used by the verifier

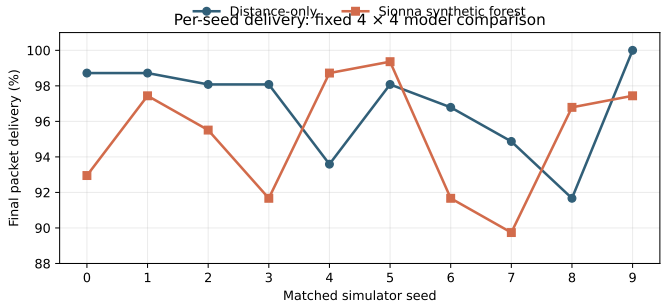


Fig. 3. Final packet-delivery rate by matched seed for the fixed 4-by-4 distance-only and Sionna-forest conditions.

coordinates and metadata. Second, a baseline demonstration run records the simulated network and fire-front evolution as an animated GIF. Third, the fixed benchmark and added channel-model/threshold studies run matched seeds (50 and 90 runs). Fourth, the expanded harness evaluates the paired size/direction/speed study described below (1,620 runs). Plotting scripts generate figures from the saved raw CSV files. Both datasets, matrices, and JSON files are included in the supplement so that the means and plotted points can be recomputed.

The open scene serves as an integration check. If its link-gain matrix does not reproduce the distance-only graph at the selected settings, the difference could reflect an unintended threshold, indexing mismatch, or channel-model choice rather than vegetation. The fact that the open scene reproduces the baseline here is a useful consistency result, but it is specific to the current grid, gain thresholds, radio range, and propagation setup. It should not be assumed for other frequencies, antenna patterns, or layouts.

X. Detailed Results and Curves

A. Per-seed packet delivery

Fig. 3 plots matched distance-only and Sionna-forest delivery for each of ten seeds. It shows the seed-to-seed spread behind the means in Table I; these are paired simulation runs, not a time trend.

B. Expanded factorial evaluation and paired statistics

To test whether the fixed 4-by-4 result generalizes across deployment scale and fire trajectories, a separate factorial experiment compared the distance-only graph with Sionna-derived forest graphs on 3-by-3, 4-by-4, and 5-by-5 grids (9, 16, and 25 nodes). Each configuration used three headings (0°, 90°, and 180°), three front speeds (1.0, 1.5, and 2.0 m/s), and 20 matched simulator seeds. Two independently generated synthetic forest layouts (scene seeds 42 and 11) were evaluated for every grid. This produces 1,620 simulator runs: for each size, trajectory, and seed, one distance-only run and two forest-channel runs. All expanded Sionna matrices use 2.4 GHz, 10 m spacing, 1.5 m node height, ray depth 1, and a -100 dB edge threshold. This is a separate exploratory protocol from the preceding fixed 4-by-4 benchmark, which used a different forest matrix, ray depth, and ten-seed comparison; their estimates are therefore reported separately.

The inferential unit is the simulator seed, not an individual direction, speed, or channel-layout run. For each seed and grid size, the distance-only outcome is averaged across the nine heading/speed combinations; the forest outcome is averaged across the same combinations and both layouts. We report paired 95% Student-t confidence intervals for mean differences and an exact two-sided sign test for directional consistency across the 20 seed clusters. Delivery-rate differences are expressed in percentage points. The factor-level CSV is descriptive; the seed-cluster summary is used for inference to avoid treating repeated settings from one seed as independent samples.

The delivery intervals in Table III all include zero. The study therefore does not establish a consistent increase or decrease in mean delivery across the tested sizes, even though the radio graph becomes sparser. Mean usable undirected links were 20 versus 18.5 for 3-by-3, 42 versus 34 for 4-by-4, and 72 versus 59 for 5-by-5 (distance-only versus forest). The 3-by-3 forest value is an average over its two layouts, which retained 18 and 19 links. This contrast shows why link count alone cannot predict delivery: the locations of removed links and the fire/event trajectory also matter.

The fixed 4-by-4 run used one heading (30°) and speed (1.5 m/s), yielding 0.096% absolute speed error. Across the

TABLE III
EXPANDED PAIRED PACKET-DELIVERY RESULTS

Grid	Distance-only (%)	Forest (%)	Paired difference (pp, 95% CI)	Sign-test p
3×3	95.34	96.37	$+1.03\ [-0.57,\ 2.64]$	0.263
4×4	96.08	95.19	$-0.89\ [-2.29,\ 0.52]$	0.824
5×5	93.46	94.18	$+0.72\ [-0.71,\ 2.16]$	0.503

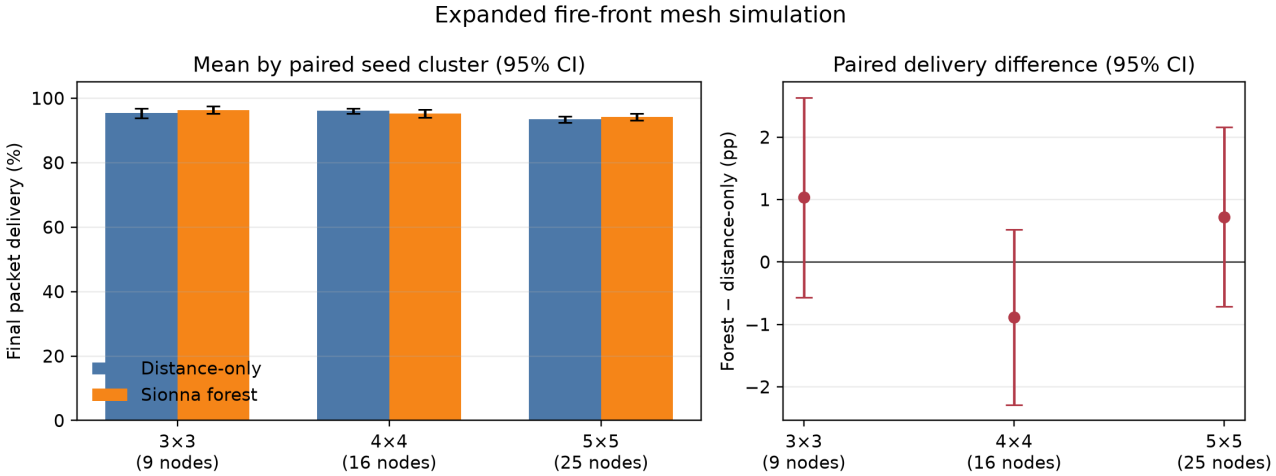


Fig. 4. Expanded packet-delivery results. Left: means and 95% confidence intervals across 20 seed-cluster means. Right: paired forest-minus-distance-only differences and 95% intervals; all three include zero.

broader factorial study, mean absolute errors were 85.19%, 48.17%, and 38.45% on the 3-by-3, 4-by-4, and 5-by-5 distance-only grids; forest means were 85.19%, 48.08%, and 38.34%. A supplementary 30-seed rerun of the 3-by-3 case (5% link dropout, 30°, 1.5 m/s) gave a signed error of +34.55%, identical across seeds, so the error is upward at least in that condition. These errors are nearly identical for the distance-only and forest graphs and belong to the baseline estimator, which this paper does not modify; we therefore do not treat them as a channel effect.

The baseline computes $\hat{v} = 1/\|\nabla \hat{T}\|$, a reciprocal that can bias speed upward under noise; we did not isolate this cause, and improving the estimator is outside the scope of this study.

C. Graph-level interpretation

The distance-only and open Sionna matrices produce the same graph under the current 18 m cap. Both statistical baselines also retain 42 links at the -100 dB cutoff. The synthetic forest loses eight links because blocked paths fall below that threshold. This confirms that the result depends on scene geometry and thresholding; it does not establish that Sionna improves delivery over calibrated statistical models.

D. Gain-threshold sensitivity

At the illustrative link-budget cutoff of -100 dB, the forest graph has 34 links. The original -105/-100/-95 dB sweep lies on a plateau. A second ten-seed sweep at -100, -72, -70, -68, -66, -64, -62, -60, and -58 dB shows the transition: the graph remains at 34 links through -64 dB, drops to 21 at -62 dB, five at -60 dB, and zero at -58 dB. Delivery means at the last three

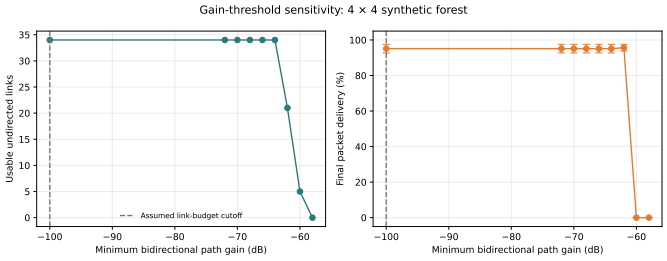


Fig. 5. Gain-threshold sensitivity with ten matched seeds. The dashed line marks the illustrative -100 dB link-budget cutoff; stricter cutoffs are stress tests.

cutoffs are 95.13%, 95.64%, and 0%. Thresholds above -100 dB are link-margin stress tests, not calibrated receiver settings; the simulator also retains an 18 m geometric range cap.

E. Wind and canopy together

The baseline simulator already models wind through an elliptical fire front whose length-to-breadth ratio follows Alexander's relation, $L/B = 1 + 0.0012\,W^{2.154}$ with W in km/h, but it does so with a distance-only radio. We therefore crossed three wind conditions ($W = 0, 20, 40$ km/h) with the distance-only and forest graphs, at 3-by-3, 4-by-4, and 5-by-5 grids, front headings of 0, 90, and 180 degrees, 1.5 m/s, 10 simulation seeds, and both forest layouts (810 runs). The unmodified simulator was used, and the gain matrix was injected at run time by replacing the neighbor-graph builder inside the harness; this setup reproduced 72 of 72 sampled rows of the earlier expanded study exactly. Wind enters only

through the baseline's fire-shape model; we did not model wind effects on the radio channel.

Wind is by far the larger effect (Table IV). At 4-by-4, distance-only delivery falls from 96.8% (calm) to 78.6% (20 km/h) and 61.4% (40 km/h), mean direction error rises from 0° to 58°, and arrival-time error rises from 6 s to 489 s. In contrast, every paired delivery interval for forest minus distance-only includes zero, in calm air and in wind alike, so we find no detectable interaction between canopy and wind in delivery. Direction, speed, and arrival errors do change when forest links are removed, but they are deterministic for a given trajectory (zero spread across seeds), their sign is not consistent across wind levels, and we do not interpret them as a systematic channel effect. The large wind errors are the baseline's straight-front estimator meeting an elliptical front, which is a property of the baseline rather than of the added channel.

F. Estimator results and scope

In the fixed comparison, direction error changes from 2.115° to 2.879° and ETA error from 6.334 s to 6.321 s. The fixed-run speed error is 0.096% for one 30°, 1.5 m/s trajectory. Table V shows the much larger mean absolute errors over the broader heading/speed set. The supplementary signed 3-by-3 check showed an upward error in that condition; the cause is not isolated here, and the estimator is not modified.

G. 3-D channel and topology views

Fig. 6 shows the two Sionna scenes, and Fig. 2 maps their gains to the resulting mesh graph. The geometry helps explain why a channel matrix is useful: the forest case includes obstacles in three dimensions, while the open case provides a clear reference. The topology plot is an analytical visualization generated from the path-gain matrices, not a screenshot of an actual field deployment. The green obstacle geometry is synthetic, and the path-gain color scale represents simulated channel output.

XI. Five Practical Questions

This section answers five practical questions about wireless mesh sensing and relates each answer to the implemented study.

1. Why use wireless mesh sensing?

A mesh lets sensor nodes forward data for one another when a direct connection to the gateway is unavailable. Multiple possible routes can also help reports continue to travel when a link or relay node is lost. This is useful in remote fire-monitoring areas where installing wired backhaul is difficult. The benefit depends on node placement, radio coverage, power, and the number of alternative routes; mesh networking does not guarantee delivery after every failure [1], [2].

2. What is a wireless mesh sensor network?

It is a group of sensor-equipped devices that both collect measurements and relay packets for neighboring devices. Measurements may travel through several intermediate nodes before reaching a gateway. In this project, temperature observations and node-event records are carried over a graph whose edges represent usable radio links. The graph can be built from distance rules or from a propagation model, such as the Sionna RT path-gain matrices evaluated here.

3. Where can it be used?

Wireless mesh sensing can support monitoring in forests, farms, remote infrastructure, and other sites where hazards must be observed but wired communication is limited. For wildfire monitoring, nodes can be placed across an area to report environmental measurements and relay alerts toward an operations gateway. A practical deployment would still require a site survey, suitable weather-resistant hardware, an adequate power plan, and radio measurements in the local terrain and vegetation. The synthetic scenes in this study are a simulation aid, not a surveyed forest.

4. How does this project use a mesh?

The supplied project baseline records node-stamped event evidence, checks whether a silent node was likely lost to fire, estimates the fire-front direction and speed from confirmed observations, and rebuilds routes after confirmed node losses. The added work supplies link gains generated from open and synthetic-forest 3-D scenes to evaluate how the radio environment changes the mesh graph. Thus, the article evaluates the communication-channel model around the existing event-processing workflow; it does not claim to introduce the baseline timestamp, verifier, estimator, or rerouting mechanisms.

5. What did the study find?

For the fixed 4-by-4 comparison, the synthetic-forest graph contained 34 usable links compared with 42 for distance-only and open-scene conditions. Its paired packet-delivery change relative to distance-only was -1.73 percentage points, with a 95% confidence interval from -4.80 to +1.34 points; the ten-seed result is therefore inconclusive about a delivery penalty. The separate 1,620-run study also found no consistent delivery change across the tested grid sizes. These synthetic results do not establish field performance or improved delivery. The channel parameters are illustrative, and the simulator uses gain-based binary links rather than a radio-specific SNR-to-PER model.

XII. Threats to Validity and Limitations

A. Propagation-model validity

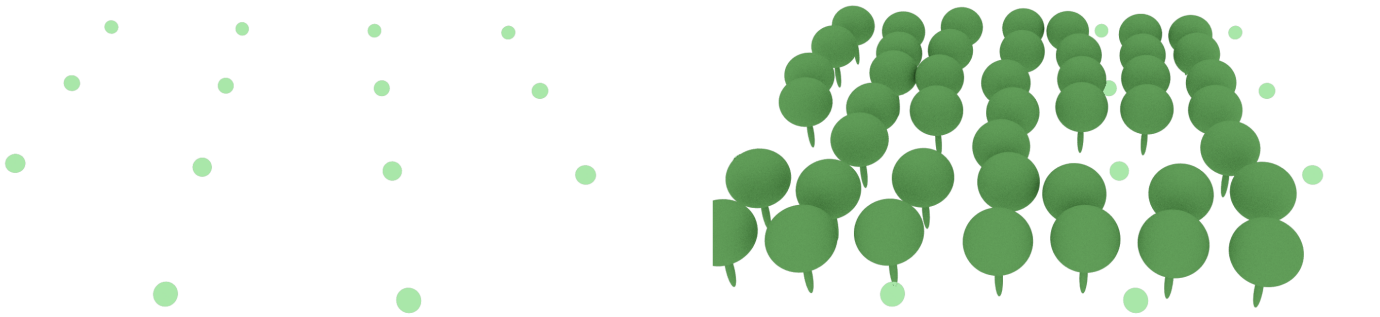
The vegetation scene uses simple trunk and canopy shapes, a single illustrative radio material, and one carrier frequency. Real vegetation is heterogeneous in species, moisture, density, season, and geometry. Sionna RT is sensitive to scene and material assumptions; therefore, the absolute link-gain

TABLE IV
WIND × CANOPY STUDY (MEAN OVER HEADINGS, SEEDS, AND LAYOUTS). LINKS AND ERROR COLUMNS SHOW DISTANCE-ONLY/FOREST; DELIVERY INTERVAL IS FOREST MINUS DISTANCE-ONLY (PP, 95% CI OVER SEED CLUSTERS).

Grid	Wind	Links	Dist. (%)	Forest (%)	Paired difference (pp)	Dir. err (°)	ETA err (s)
3×3	Calm (0)	20/18.5	95.7	95.8	$+0.09\ [-3.47,\ +3.64]$	0.0/0.0	9/9
3×3	20 km/h	20/18.5	68.2	67.1	$-1.07\ [-3.92,\ +1.79]$	15.5/10.0	63/63
3×3	40 km/h	20/18.5	21.6	21.6	$+0.01\ [-0.53,\ +0.54]$	62.6/56.9	401/401
4×4	Calm (0)	42/34.0	96.8	96.1	$-0.67\ [-2.76,\ +1.42]$	0.0/0.2	6/6
4×4	20 km/h	42/34.0	78.6	77.5	$-1.12\ [-2.44,\ +0.19]$	1.3/11.4	72/72
4×4	40 km/h	42/34.0	61.4	61.3	$-0.10\ [-0.67,\ +0.47]$	58.0/40.8	489/487
5×5	Calm (0)	72/59.0	93.3	94.6	$+1.26\ [-0.59,\ +3.12]$	0.0/0.3	6/7
5×5	20 km/h	72/59.0	63.2	63.9	$+0.69\ [-0.52,\ +1.90]$	9.6/17.5	100/100
5×5	40 km/h	72/59.0	35.8	35.7	$-0.15\ [-0.59,\ +0.29]$	52.2/39.5	480/480

TABLE V
DESCRIPTIVE ESTIMATOR ERRORS ACROSS THE EXPANDED TRAJECTORY SET. VALUES AVERAGE HEADINGS, SPEEDS, FOREST LAYOUTS, AND 20 SEEDS; THE SIMULATOR IS DETERMINISTIC FOR SEVERAL METRICS.

	Distance-only			Sionna forest		
Grid	Direction error (°)	Speed error (%)	ETA error (s)	Direction error (°)	Speed error (%)	ETA error (s)
3×3	0.000	85.185	10.256	0.000	85.185	10.256
4×4	0.000	48.168	10.232	0.162	48.076	8.449
5×5	0.000	38.450	11.215	0.183	38.335	11.288



(a) Open reference scene (b) Synthetic forest scene

Fig. 6. Sionna RT radio scenes used to generate the open and forest link-gain matrices. The forest geometry is a simplified, deterministic obstacle model.

values and derived edge set should be treated as synthetic until compared with measured links. A realistic next step would import a surveyed terrain/vegetation model, characterize antenna patterns, calibrate material parameters, and validate path gain with an independent measurement set.

B. Static channel and node failures

The generated matrix is fixed for a run. It does not model temporal fading, moving branches, fire-related structural changes, dynamic smoke and moisture, interference, duty cycling, or battery depletion. Node death is prescribed by the existing fire simulation and is not a hardware survivability test. Correlated losses can be more severe than independent missing edges: one fire front may remove several relays in sequence. Future tests should vary link state over time and include relay failures correlated with front location.

C. Statistical scope

The original ten-seed benchmark uses one 4-by-4 topology, one spacing, one frequency, one channel realization, and one prescribed fire trajectory; it is a compact reproducibility check, not a high-confidence estimate of deployment performance. The expanded study adds 3-by-3 and 5-by-5 grids, three headings, three speeds, two forest layouts per grid, and paired seed-cluster intervals. It still uses one node spacing and frequency and a small set of synthetic layouts. Repeated rows from the same seed or repeated thresholds on one matrix must not be counted as independent field samples. The delivery confidence intervals include zero for all three grid sizes. More channel geometries, irregular placements, time-varying links, fire-model variability, and measured radio/fire data are needed before making broader claims.

D. Clock and estimator assumptions

The timestamp strategy assumes that a node-local timebase is monotonic and that event timestamps can be related to a common simulation epoch. In physical devices, clock drift, resets, oscillator tolerance, and synchronization messages need explicit treatment. The project includes configurable clock-jitter experiments, but the Sionna channel comparison reported here does not sweep clock jitter. Similarly, the arrival-time plane is a local approximation and can be poorly conditioned when confirmed deaths are clustered or too few. An operational design should report estimator confidence, reject ill-conditioned fits, and consider curved or piecewise models when the fire front is not planar.

XIII. Reproducibility Package and Next Experiments

The reproducibility package contains the Sionna scene generator, statistical-baseline generator, fixed model-comparison and threshold-sweep runners, signed-speed and last-gasp ablation runner, expanded paired-study runner, channel matrices and metadata, raw run tables, and figures. The original project archive remains untouched. The estimator and verifier are existing project mechanisms; this paper does not claim to have added or validated them.

Next steps are to calibrate vegetation and antenna materials against measured links, select a specific transceiver and derive its link budget, and replace the binary gain cutoff with a PHY-specific SNR-to-PER curve. The project does not identify a radio, so imposing an assumed PER curve would create unsupported precision. A hardware-oriented study should also vary temporal fading, clock drift, and last-gasp loss, and directly ablate node-stamped last gasp, heartbeat fallback, and gateway-confirmation time.

For an eventual field study, deploy a small number of weatherproof sensor nodes with calibrated temperature sensors and synchronized clocks. Measure RSSI or link-quality indicators across representative vegetation and weather conditions; deliberately interrupt selected paths; compare node-stamped last-gasp events against a reference clock; and annotate true fire/no-fire loss states in a controlled thermal test. Such data would enable calibration of both the radio scene and the verifier and provide a defensible path from simulation to an operational trial.

XIV. Conclusion

The contribution of this paper is a Sionna RT channel integration and a reproducible comparison with distance-only and statistical propagation baselines. It does not introduce the baseline last-gasp, timestamp, verification, estimator, or rerouting mechanisms. The forest scene changes the graph from 42 to 34 links, but the ten-seed paired delivery interval is wide and the post-hoc equivalence check is inconclusive. The 1,620-run study likewise found no consistent delivery change across grid sizes, while its broad trajectory set exposed large speed-estimation errors. Crossing the forest with the baseline's wind model showed that wind, not canopy, dominates the degradation in delivery and in direction and arrival-time error, with no detectable interaction. Before operational conclusions, the scene needs calibration against measured links and the radio PHY needs a specified SNR-to-PER model. Further work should test physical clock behavior and compare node-stamped last gasp, heartbeat fallback, and gateway confirmation time under the scene-derived channel.

Appendix A
Reproduction Checklist

The fixed-layout benchmark uses the recorded grid, spacing, height, frequency, seed, and ray depth. Its harness runs ten seeds at -95, -100, and -105 dB and produces 50 rows.

The expanded study uses six saved forest matrices for 3-by-3, 4-by-4, and 5-by-5 grids, with layout seeds 42 and 11. It tests three headings, three speeds, 20 matched simulation seeds, ray depth 1, and a -100 dB threshold. Run `run_expanded_study.py` with the separate simulator checkout specified by `--project-root`, the matrix folder specified by `--channel-root`, and a destination specified by `--outdir`. The run produces 1,620 rows. Recreate the confidence-interval figures from `expanded_raw.csv`, `seed_cluster_means.csv`, and `expanded_statistics.csv`. Factor-level results are in `factor_statistics.csv`. Minor package-version differences may change rendered figures but should not change the experiment settings.

The targeted signed-speed check uses the 3-by-3 grid, 30° fire heading, 1.5 m/s true speed, and 5% temporary-link dropout for 30 seeds. The matched last-gasp ablation uses a 4-by-4 grid and 30 seeds with only the final telemetry packet suppressed in the ablation arm. Run `run_article_revision_experiments.py`; its per-seed and summary files are listed in the supplement.

Appendix B
Supplementary Checks

Two checks use the baseline unchanged and are reported only as supplements. (i) A 30-seed 3-by-3 rerun gave a signed speed error of +34.55%, identical across seeds. (ii) A matched 30-seed 4-by-4 ablation suppressed only `LAST_GASP` packets at the sender boundary: 13 versus 12 true fire deaths were confirmed per run and mean absolute event-time error on common deaths was 0.336 s versus 0.864 s, with zero false positives in both arms. Delivery differed by +0.087 pp (95% CI [-0.067, +0.241]). Death counts and timestamps were identical across seeds, so these are deterministic outcomes for this setup, and the ablation does not exercise the separate `had_last_gasp` verifier flag or the forest channel.

The wind × canopy study is reproduced by `wind_forest.py` (run-time injection of the gain matrix without editing the simulator), `run_wind.py`, and `analyze.py`; outputs are `wind_forest_raw.csv` and `wind_forest_summary.csv`.

Appendix C
Artifact Inventory

The supplement contains the original open and forest channel matrices, node-coordinate files, scene metadata, three-

dimensional renders, topology plot, fixed benchmark results (`trees_comparison.csv`), and baseline animation (`baseline_demo.gif`). The expanded experiment adds six Sionna matrices for two forest layouts at each grid size, the raw 1,620-run data (`expanded_raw.csv`), seed-cluster means, factor-level summaries, statistical report, metadata, and confidence-interval plots. These files let a reviewer inspect matrix ordering, reproduce the statistics, and distinguish the fixed benchmark from the expanded factorial study. The signed-speed and last-gasp message ablations add `signed_speed_3x3_raw.csv`, `signed_speed_3x3_summary.csv`, `last_gasp_ablation_raw.csv`, and `last_gasp_ablation_summary.csv`, generated by `run_article_revision_experiments.py`. The original project core remains unchanged.

APPENDIX D
MESSAGE TIMELINE AND EVIDENCE TRACEABILITY

A node loss has several relevant times. t_{HB} is the most recent node-stamped heartbeat, t_{LG} is the node-stamped last gasp, t_{rx} is gateway receive time, and t_{conf} is the later verifier decision. Only t_{LG}, or t_{HB} when no last gasp arrives, represents a candidate event time. Substituting t_{conf} would add silence, routing, and verification delays to the inferred fire-arrival time.

$$\hat{t}_{event} = \begin{cases} t_{LG} - t_0, & \text{if a valid last-gasp stamp is available,} \\ t_{HB} - t_0, & \text{otherwise, using the last node-stamped heartbeat,} \end{cases} \tag{4}$$

Here t_0 is the adapter's local-to-relative-time reference. A heartbeat can predate the thermal transition, so it is a less precise fallback than a received last gasp. A deployment must also measure clock drift and last-gasp delivery probability.

The evidence supports software and synthetic-scene claims within the tested setup. It does not establish field performance.

1) **Event-time precedence.** An adapter test compares a node-stamped last gasp with delayed gateway confirmation. It checks software timestamp selection, not physical clock accuracy.
2) **Heartbeat fallback.** A test removes the last gasp and checks the node-stamped heartbeat fallback. Field last-gasp loss probability remains unmeasured.
3) **False-death filtering.** Project scenarios exercise neighbor corroboration, thermal evidence, and non-fire exclusions. Correlated field outages need further study.
4) **Route repair.** The simulator records a 100 ms mean repair delay for this run set; hardware timing is unmeasured.
5) **Radio integration.** Saved Sionna matrices, coordinates, metadata, and scene renders document the synthetic channel input and its node ordering.
6) **Topology change.** The saved comparison shows 42 open links and 34 forest links for this 4-by-4, 2.4 GHz setup and its selected scene parameters.
7) **Baseline false positives.** One animation run reports 13 confirmed fire deaths among 16 nodes and zero false positives. One run is not a broad false-alarm estimate.
8) **Last-gasp message ablation.** A matched 30-seed simulation suppressed the final telemetry packet and measured confirmed deaths, misses, timestamp error, and delivery. It does not exercise the separate external `had_last_gasp` verifier flag or establish hardware effects.

Physical clock drift, calibrated vegetation loss, and end-to-end fire/no-fire labels require separate measurements before operational claims.

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